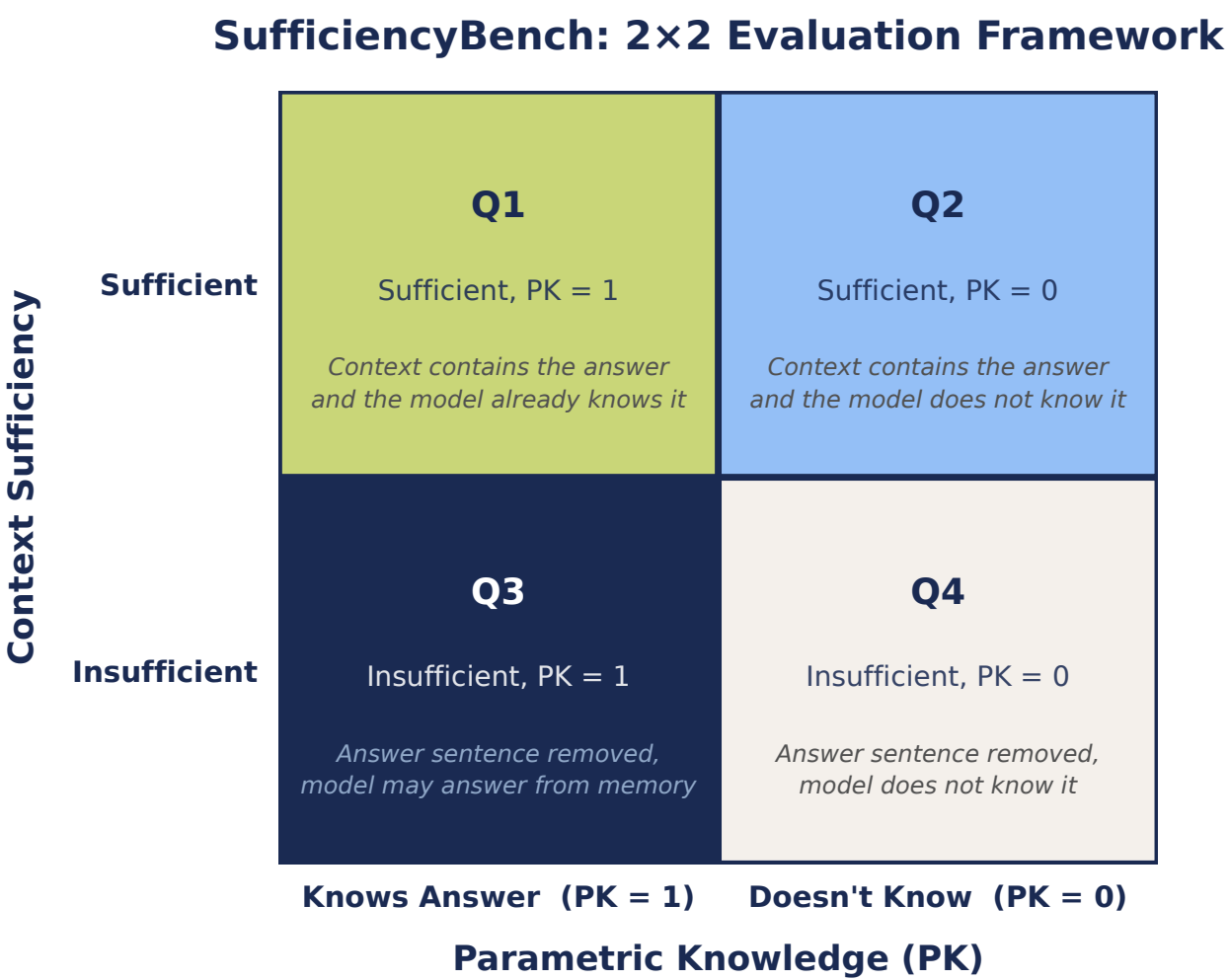
SufficiencyBench: 2×2 Evaluation Framework
Context Sufficiency
Sufficient
Insufficient
Q1
Sufficient, PK = 1
Context contains the answer and the model already knows it
Q2
Sufficient, PK = 0
Context contains the answer and the model does not know it
Q3
Insufficient, PK = 1
Answer sentence removed, model may answer from memory
Q4
Insufficient, PK = 0
Answer sentence removed, model does not know it
Knows Answer (PK = 1)
Doesn't Know (PK = 0)
Parametric Knowledge (PK)